

#HW 12

● Graded

Student

Pamela Gutierrez

Total Points

93 / 100 pts

Question 1

(no title)

30 / 30 pts

✓ - 0 pts Correct

Question 2

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 1 pt minimum theoretical power input value?

- 1 pt wrong T_c and T_h temp. taken

- 1 pt is 3 hp sufficient or not?

Question 3

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 3 pts reversible heat pump calculation?

- 5 pts incomplete solution

Question 4

(no title)

11 / 15 pts

- 0 pts Correct

- 2 pts Pressure at end of isothermal expansion = 76.1 kPa

✓ - 4 pts pressure not calculated

Question 5

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 2 pts Case (a) is irreversible

Question 6

(no title)

12 / 15 pts

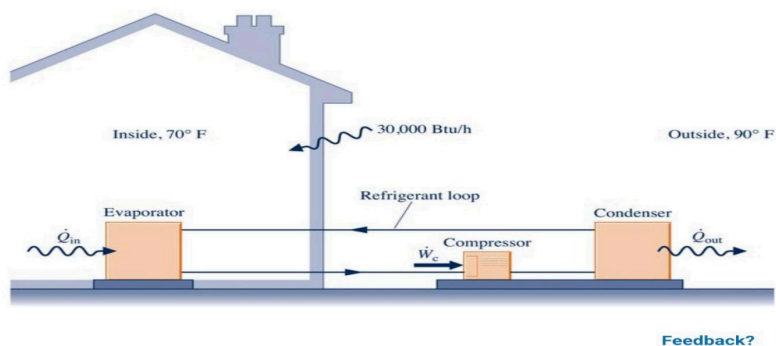
– 0 pts Correct

✓ – 3 pts incomplete solution

Question assigned to the following page: [1](#)



- (a) As shown in the figure below, an air conditioner operating at steady state maintains a dwelling at 70°F on a day when the outside temperature is 90°F. If the rate of heat transfer into the dwelling through the walls and roof is 30,000 Btu/h, might a net power input to the air conditioner compressor of 3 hp be sufficient? If yes, determine the coefficient of performance. If no, determine the minimum theoretical power input, in hp.



$$T_L = 70^\circ\text{F} \quad 70 + 459.67 = 529.67^\circ\text{R}$$

$$T_H = 90^\circ\text{F} \quad 90 + 459.67 = 549.67^\circ\text{R}$$

$$\text{COP} \quad \text{COP}_{\text{max}} = \frac{T_L}{T_H - T_L}$$

$$\text{COP} = \frac{549.67^\circ\text{R}}{549.67 - 529.67} = 26.48$$

$$\text{COP} = \frac{\dot{Q}_{\text{in}}}{W_c} \quad W_c = \frac{30,000 \text{ Btu/h}}{26.48} = 1133 \text{ Btu/h} \cdot \frac{1 \text{ hp}}{2544.43} = 0.45 \text{ hp}$$

Yes.

Questions assigned to the following page: [1](#) and [2](#)



EXERCISE

6.17.47: (Problem 5.47 in the 9th edition).



- (a) To maintain a dwelling steadily at 68°F on a day when the outside temperature is 32°F, heating must be provided at an average rate of 740 Btu/min. Compare the electrical power required, in kW, to deliver the heating using (a) electrical-resistance heating, (b) a heat pump whose coefficient of performance is 3.0, (c) a reversible heat pump operating between hot and cold reservoirs at 68°F and 32°F, respectively.

[Feedback?](#)

a) electrical-resistance heating

$$\dot{W}_a = \frac{\dot{Q}_{in}}{E_{eff}} = \frac{740 \text{ Btu/min}}{1} = 740 \text{ Btu/min} \cdot \frac{1 \text{ kW}}{56.87 \text{ Btu/min}} = 13 \text{ kW}$$

b) Heat pump

$$\text{COP} = \frac{\dot{Q}}{W} \quad W = \frac{Q}{\text{COP}} \quad W_b = \frac{740 \text{ Btu/min}}{3} \cdot \frac{1 \text{ kW}}{56.87 \text{ Btu/min}} = 4.34 \text{ kW}$$

c) reversible pump

$$T_H = 68^\circ\text{F} + 459.67 = 527.67 \text{ R}$$

$$T_C = 32^\circ\text{F} + 459.67 = 491.67 \text{ R}$$

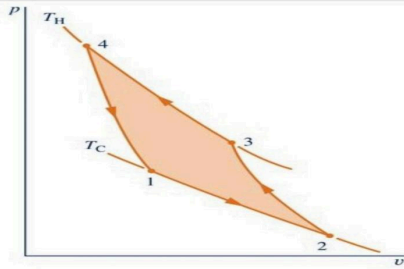
$$\text{COP} = \frac{T_H}{T_H - T_C} = \frac{527.67}{527.67 - 491.67} = 14.66$$

$$\frac{740 \text{ Btu/min}}{14.66} = \frac{50.48}{56.87}$$

$$= 0.89 \text{ kW}$$

Questions assigned to the following page: [2](#) and [3](#)

- (a) Air within a piston-cylinder assembly executes a Carnot heat pump cycle, as shown in the figure below. For the cycle, $T_H = 600$ K and $T_C = 300$ K. The energy rejected by heat transfer at 600 K has a magnitude of 250 kJ per kg of air. The pressure at the start of the isothermal expansion is 325 kPa. Assuming the ideal gas model for the air, determine (a) the magnitude of the net work input, in kJ per kg of air, and (b) the pressure at the end of the isothermal expansion, in kPa.



Feedback?

a) mag net work

$$\text{COP} = \frac{T_H}{T_H - T_C} = \frac{600 \text{ K}}{600 \text{ K} - 300 \text{ K}} = 2$$

$$\frac{250 \text{ kJ/kg}}{2} = 125 \text{ kJ/kg}$$

b) pressure

$$\frac{Q_C}{Q_H} = \frac{T_C}{T_H}$$

$$Q_C = Q_H \frac{T_C}{T_H}$$

$$= 250 \cdot \frac{300}{600}$$

$$= 125 \text{ kJ/kg}$$

Questions assigned to the following page: [3](#) and [4](#)



EXERCISE

6.17.58: (Problem 5.58 in the 9th edition).



- (a) The steady-state data listed below are claimed for a power cycle operating between hot and cold reservoirs at 1200 K and 400 K, respectively. For each case, evaluate the net power developed by the cycle, in kW, and the thermal efficiency. Also in each case apply Eq. 6.11b on a time-rate basis to determine whether the cycle operates reversibly, operates irreversibly, or is impossible.

a. $\dot{Q}_H = 600 \text{ kW}$, $\dot{Q}_C = 400 \text{ kW}$

b. $\dot{Q}_H = 600 \text{ kW}$, $\dot{Q}_C = 0 \text{ kW}$

c. $\dot{Q}_H = 600 \text{ kW}$, $\dot{Q}_C = 200 \text{ kW}$

Feedback?

$$\begin{aligned} \text{a) } W_{\text{cycle}} &= Q_H - Q_C = 600 \text{ kW} - 400 \text{ kW} \\ &= \underline{200 \text{ kW}} \end{aligned}$$

$$\eta = \frac{200}{600} = \underline{0.33} \quad \text{irreversible}$$

$$\frac{600 \text{ kW}}{1200 \text{ K}} + \frac{-400 \text{ kW}}{400 \text{ K}} = -0.5$$

$$\begin{aligned} \text{b) } W_{\text{cycle}} &= Q_H - Q_C = 600 \text{ kW} - 0 \text{ kW} \\ &= 600 \text{ kW} \end{aligned}$$

$$\eta = \frac{600 \text{ kW}}{600 \text{ kW}} = \underline{1.0}$$

$$\frac{600 \text{ kW}}{1200} + \frac{0 \text{ kW}}{400 \text{ K}} = 0.5$$

impossible

$$\text{c) } 600 - 200 = \underline{400 \text{ kW}} \quad \frac{400}{600} = \underline{0.667}$$

$$\frac{600}{1200} + \frac{-200}{400} = 0$$

reversible

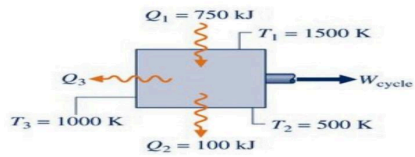
Questions assigned to the following page: [5](#) and [6](#)

**EXERCISE**

6.17.63: (Problem 5.63 in the 9th edition).



- (a) As shown in the figure below, a system executes a power cycle while receiving 750 kJ by heat transfer at a temperature of 1500 K and discharging 100 kJ by heat transfer at a temperature of 500 K. Another heat transfer from the system occurs at a temperature of 1000 K. Using Eq. 6.6a, plot the thermal efficiency of the cycle versus σ_{cycle} , in kJ/K.

[Feedback?](#)